

Backtesting Probability of Default Models

On the Reliability of the Multi-Period Normal Test

Andrija Djurovic

www.linkedin.com/in/andrija-djurovic

Internal Ratings-Based Modeling Process

- The development of an Internal Ratings-Based (IRB) model consists of two main areas: the development of the risk differentiation function and risk quantification.
- Development of the risk differentiation function is the first step in the overall model development process. The main goal of this step is to ensure the model can adequately rank risk, providing meaningful differentiation between low- and high-risk obligors or facilities.
- After developing the risk differentiation function, practitioners perform the risk quantification process, which essentially involves assigning an appropriate risk level to the output of the risk differentiation function.
- Risk quantification typically consists of specific steps such as: assignment of the base or best estimate, quantification of appropriate adjustments (AA), quantification of the margin of conservatism (MoC), and quantification of any other additional conservatism, for instance that required by certain local regulators. The first two steps are typically referred to as calibration.
- These steps apply to all three risk parameters: Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD).

Probability of Default Backtesting

- As risk quantification is a regulatory requirement, practitioners pay significant attention to complying with these requirements.
- In addition to adhering to the prescribed risk quantification process, practitioners are required to assess the appropriateness of the model estimates. This is typically referred to as backtesting.
- The main purpose of backtesting in this context is to ensure that the model estimates adequately predict the occurrence of defaults, i.e., that PD estimates constitute reliable forecasts of default rates.
- In practice, this is one of the most comprehensive validation exercises, typically investigating model estimates across different portfolio subsegments and time dimensions.
- A commonly applied approach is to backtest estimates over a recent period, usually consisting of the latest three to five years. Sometimes, this analysis is extended further back, but still with a fairly limited number of observations.

Probability of Default Backtesting cont.

- One of the approaches used to simultaneously test the accuracy of the estimates compared to recently observed defaults is the normal test. To the best of the authors' knowledge, this is the only multi-period test proposed for this purpose that simultaneously considers the difference between the estimates and observed values.
- Although common in practice, given the limited number of observations, it is questionable how reliable this test is for the intended purpose.
- The following slides provide basic details about the multi-period normal test and present the results of a simulation study that investigates the Type I error of this test under different data-generating processes for default rates.
- As the results suggest, practitioners should be clearly aware of possible test outcomes when the test is applied to a limited number of observations, and even more so in the presence of autocorrelation.
- Further details on the multi-period normal test can be found in this [study](#), where the authors also discuss several limitations of the test and emphasize the need for caution in its current formulation.

Multi-Period Normal Test

Test Statistics

The multi-period normal test simultaneously evaluates the difference between estimated and observed values over time. The test statistic is defined as follows:

$$Z_{nt} = \frac{\sum_{i=1}^T (ODR_i - PD_i)}{\sqrt{T} \cdot se}$$

where:

- PD is the calibrated PD;
- ODR is the observed default rate;
- T is the number of years;
- se is the standard error defined as:

$$\sqrt{\frac{1}{T-1} \left(\sum_{i=1}^T (ODR_i - PD_i)^2 - \frac{\left(\sum_{i=1}^T (ODR_i - PD_i) \right)^2}{T} \right)}$$

Distributional Assumptions

The Z_{nt} test statistic is assumed to follow the standard normal distribution, and the corresponding p-value is calculated accordingly. This test is used to assess accuracy as well as potential underestimation of the estimates, which directly influences the p-value calculation.

Simulation Study

This simulation investigates the Type I error of the multi-period normal test under three different default rate generating processes. Specifically, the simulation relies on the following processes:

- 1 Vasicek distribution, uncorrelated over time;
- 2 Vasicek distribution, correlated over time;
- 3 Normal approximation of the binomial distribution, uncorrelated over time.

The following steps describe the simulation procedure, assuming that default rates are generated from the Vasicek distribution:

- 1 Define the parameters of the Vasicek distribution - average PD (PD = 5%) and asset correlation ($\rho = 0.10$).
- 2 Define the sample size, T, ranging from three to ten, representing the number of years used for backtesting the PD estimates. For comparison purposes, the simulation also considers T = 100.
- 3 Define the first-order autocorrelation coefficient ($\phi = 0.30$). For the uncorrelated data-generating process, the parameter (ϕ) is set to 0.
- 4 Based on the Vasicek distribution, simulate (T) observations (under both correlated and uncorrelated assumptions) of the conditional default rate given the simulated systemic factor of the following form:

$$z_{t,t \leq T} = \phi z_{t-1} + \sqrt{1 - \phi^2} \epsilon_t$$

where $z_{t,t=1}$ and ϵ_t are drawn from the standard normal distribution.

- 5 Perform the multi-period normal test and store the resulting p-value.
- 6 Repeat Steps 4 and 5, 10,000 times.
- 7 For the selected significance level (10%), calculate the Type I error as the number of simulations with a p-value less than 10%.

Simulation Study cont.

The following steps describe the simulation procedure, assuming that default rates are generated from the normal distribution. To ensure comparability, all common parameters are kept the same as in the Vasicek distribution data-generating process.

- 1 Define the average PD (PD = 5%).
- 2 Define the sample size, T, ranging from three to ten, representing the number of years used for backtesting the PD estimates. For comparison purposes, the simulation also considers T = 100.
- 3 Based on the defined PD and T, calculate the standard deviation as $sd = \sqrt{\frac{PD(1-PD)}{T}}$.
- 4 Simulate T default rates from the normal distribution with mean PD and standard deviation sd. If any simulated values fall outside the [0, 1] range, floor them at 0 and cap them at 1.
- 5 Perform the multi-period normal test and store the resulting p-value.
- 6 Repeat Steps 4 and 5, 10,000 times.
- 7 For the selected significance level (10%), calculate the Type I error as the number of simulations with a p-value less than 10%.

Notes

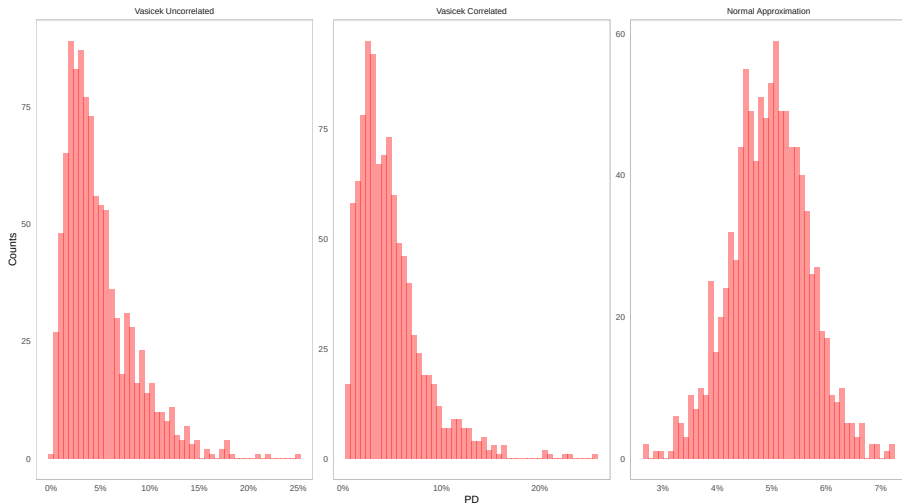
Under the null hypothesis and a correctly specified test statistic, the p-value should follow a uniform distribution with a minimum of 0 and a maximum of 1. Therefore, the expected Type I error should be approximately equal to the selected significance level.

A comparison between the simulation results assuming a Vasicek distribution with uncorrelated conditional default rates and those obtained from default rates simulated from a normal distribution illustrates the impact of potential deviations from normality in the underlying distributions.

In contrast, simulations based on Vasicek-generated correlated default rates demonstrate the impact of correlation between default rates and the appropriateness of the test, even under a relatively mild level of correlation.

Simulation Results

Distribution of Simulated PDs per Underlying Model



Simulation Results cont.

The following table presents the Type I error of the multi-period normal test for a 10% significance level, across different sample sizes (T) and underlying data-generating processes of defaults.

##	T	vasicek_uncorrelated	vasicek_correlated	normal_approx_binomial
##	3	29.61%	38.46%	22.23%
##	4	25.13%	35.88%	21.16%
##	5	21.65%	32.44%	18.74%
##	6	19.61%	32.11%	17.74%
##	7	18.89%	30.06%	16.64%
##	8	18.07%	28.53%	16.23%
##	9	17.11%	28.41%	16.10%
##	10	16.72%	27.42%	15.56%
##	100	10.32%	22.93%	10.30%

Notes

As shown in the simulation results, the deviation from normality has a decreasing impact on the Type I error as the sample size T increases. However, even a mild autocorrelation among the simulated default rates significantly affects the test results, even for a sample size of 100.

Therefore, given the primary role of this test in PD model validation, practitioners should exercise caution in relying on its outcomes, as the test's validity may be limited for evaluating this particular aspect of the model. Similar cautions have also been noted by the authors of the test.